

HAUTE: Harmonizing Action Units with Temporal-contextual Embeddings for Deepfake Detection

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Abstract

The proliferation of highly realistic deepfake videos threatens public trust and the integrity of digital information. However, detecting sophisticated deepfakes requires analysis beyond surface-level visual artifacts. We propose Harmonizing Action Units with Temporal-contextual Embeddings (HAUTE), integrating physiological muscle dynamics with holistic semantic context through adaptive attention mechanisms. HAUTE captures temporal Action Unit coordination patterns and high-level contextual embeddings, enabling the model to reveal synthesis-induced inconsistencies imperceptible to isolated modalities. Extensive experiments demonstrate state-of-the-art performance with strong cross-dataset adaptability, particularly on commercial tool-based high-quality deepfakes, advancing trustworthy content verification for web ecosystems.

CCS Concepts

• **Computing methodologies** → **Computer vision**; **Artificial intelligence**; **Computer vision tasks**; *Image representations*.

Keywords

Deepfake Detection, Action Unit, Feature Fusion, Temporal Dynamics

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1 Introduction

The rapid advancement of generative AI technologies has enabled even non-technical users to create and disseminate highly realistic deepfake videos easily. As massive volumes of sophisticated deepfake videos proliferate across social media and online platforms, such content poses serious social harms, including misinformation propagation, political manipulation, and the production of non-consensual pornography. This accelerates the erosion of trust in web-based content, fundamentally threatening the verifiability of digital media authenticity and the credibility of online information ecosystems. In response to these threats, extensive research has been conducted on deepfake detection, such as global

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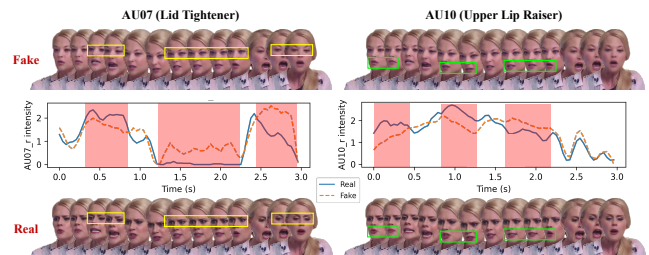


Figure 1: Physiological coordination anomalies in deepfake facial expressions. AU intensity trajectories for AU07 (Lid Tightener) and AU10 (Upper Lip Raiser) extracted from real and deepfake videos. Colored boxes indicate tracked facial regions. Red-shaded intervals highlight temporal asynchronies and intensity discrepancies in deepfakes.

appearance-based approaches [14] and frequency-domain methods [18]. However, recent deepfake generation models effectively suppress low-level visual artifacts, posing fundamental limitations for artifact-based detection approaches [16].

We shift focus to a more fundamental cue that is resistant to forgery: the physiological dynamics of facial musculature. In authentic human facial expressions, specific muscle units, i.e., Action Unit (AU) as defined in the Facial Action Coding System (FACS), activate in physiologically coordinated patterns during expression transitions [4]. For example, AU07 (Lid Tightener) and AU10 (Upper Lip Raiser) activate in temporal synchrony when expressing disgust. However, such inter-muscle coordination is difficult to replicate accurately during deepfake synthesis, resulting in subtle asynchronies such as delayed activation or intensity mismatches in AU07 (see Figure 1) [8]. Thus, synthesized faces exhibit physiologically unnatural patterns across both temporal AU dynamics and holistic facial movement coordination.

Motivated by this observation, several AU-based deepfake detection methods have been proposed [1, 13]. However, key limitations remain unaddressed: (1) relying solely on AU intensity signals may overlook the broader semantic context of the face, and (2) modeling only static inter-AU relationships may insufficiently capture dynamic changes over time.

To address these limitations, we propose Harmonizing Action Units with Temporal-contextual Embeddings (HAUTE), which integrates physiological muscle dynamics with global contextual information through three specialized components. The Temporal Physiological Encoding Stream models bidirectional temporal patterns of muscle activation, while the Holistic Semantic Encoding Stream captures high-level facial semantics via pretrained vision encoders. The Attentive Feature Harmonization Module adaptively

fuses these representations, enabling the integration of local physiological patterns with global semantic context to reveal synthesis artifacts imperceptible to individual modalities.

Our contributions are twofold: (1) a synergistic integration of physiological dynamics and contextual semantics through adaptive attention mechanisms for robust deepfake detection, and (2) comprehensive empirical validation demonstrating superior performance and cross-dataset adaptability. HAUTE is expected to advance visual content verification for web ecosystems, contributing to the restoration of digital trust.

2 Method

2.1 Problem Formulation

To establish a principled framework for deepfake detection that integrates physiological and contextual cues, the detection task is formulated as follows. Given an input video $\mathcal{V}$ consisting of T frames, the goal is to learn a classification function $\mathcal{F} : \mathcal{V} \rightarrow \{0, 1\}$ that determines whether $\mathcal{V}$ is real or fake (i.e., 0 or 1).

Each frame f_t yields two complementary representations for deepfake detection.

Physiological Dynamics Representation Frame-level AU intensities are extracted from the input video using the OpenFace 2.0 toolkit. OpenFace provides FACS-based AU analysis with activation intensities as values in the range $[0, 5]$ for each AU. To capture the inter-muscle coordination anomalies in deepfake, seven AUs associated with major facial expressions are utilized: AU01 (Inner Brow Raiser), AU04 (Brow Lowerer), AU07 (Lid Tightener), AU10 (Upper Lip Raiser), AU12 (Lip Corner Puller), AU14 (Dimpler), and AU26 (Jaw Drop) [7]. An AU intensity sequence $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_T] \in \mathbb{R}^{T \times K}$ is extracted, where $K = 7$ denotes the number of Action Units and $\mathbf{a}_t \in \mathbb{R}^K$ represents the AU activation vector at time t .

Semantic Contextual Representation A frame sequence $\mathbf{F} = \{f_1, f_2, \dots, f_T\}$ temporally aligned with the AU intensity sequence is extracted from the input video $\mathcal{V}$. To ensure consistent processing, both physiological and contextual representations are normalized to a fixed length $T_{\max}$ via zero-padding (for $\mathbf{A}$) or last-frame repetition (for $\mathbf{F}$) when shorter than $T_{\max}$, and truncation for longer sequences.

Finally, HAUTE takes the AU intensity sequence $\mathbf{A}$ and frame sequence $\mathbf{F}$ as input to produce a probability $p \in [0, 1]$ via sigmoid-activated logit, where $p > \tau$ indicates a deepfake prediction for threshold τ .

2.2 HAUTE: Harmonizing Action Units with Temporal-contextual Embeddings

HAUTE integrates physiological and contextual cues through three core components: (1) Temporal Physiological Encoding Stream captures temporal patterns of facial muscle dynamics, (2) Holistic Semantic Encoding Stream extracts high-level contextual embeddings from facial frames, and (3) Attentive Feature Harmonization Module adaptively fuses both representations to exploit their complementary information.

2.2.1 Temporal Physiological Encoding Stream The Temporal Physiological Encoding Stream processes the AU sequence $\mathbf{A}$ to capture temporal muscle dynamics that reveal inter-AU coordination anomalies in deepfakes. A bidirectional LSTM encodes both historical AU activation trajectories—how muscles have moved up

to time t —and future evolution patterns by incorporating subsequent movements. At each timestep t , the forward and backward LSTM cells compute:

$$h_t^{AU} = \left[\overrightarrow{\text{LSTM}}(\mathbf{a}_t, \overrightarrow{h_{t-1}}; \theta_{fwd}); \overleftarrow{\text{LSTM}}(\mathbf{a}_t, \overleftarrow{h_{t+1}}; \theta_{bwd}) \right] \quad (1)$$

where $\overrightarrow{h_{t-1}}$ and $\overleftarrow{h_{t+1}}$ denote the hidden states from forward and backward passes respectively, θ_{fwd} and θ_{bwd} are learnable parameters, and $h_t^{AU} \in \mathbb{R}^D$ is the concatenated hidden representation at timestep t with shared embedding dimension D . This yields the physiological representation $\mathbf{H}^{AU} = \{h_1^{AU}, \dots, h_T^{AU}\} \in \mathbb{R}^{T \times D}$. Incorporating bidirectional temporal context enables detection of subtle asynchronies—physiological activation delays and intensity mismatches—imperceptible in isolated frames. $\mathbf{H}^{AU}$ thus captures physiological coordination patterns that distinguish genuine faces from deepfakes exhibiting synthesis-induced anomalies.

2.2.2 Holistic Semantic Encoding Stream In addition to fine-grained muscle dynamics captured by AU intensities, holistic facial cues, including lighting, pose consistency, and semantic coherence, are essential for comprehensive deepfake detection. The Holistic Semantic Encoding Stream extracts such high-level contextual features via a pretrained vision encoder followed by a learnable projection:

$$\mathbf{v}_t = \text{Encoder}(f_t; \theta_{\text{enc}}) \in \mathbb{R}^{D_{\text{enc}}} \quad (2)$$

$$h_t^{\text{SEM}} = \mathbf{W}_{\text{proj}} \mathbf{v}_t + \mathbf{b}_{\text{proj}} \in \mathbb{R}^D \quad (3)$$

where θ_{enc} denotes the frozen encoder parameters, D_{enc} is the output dimension of the encoder, and $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{D \times D_{\text{enc}}}$ and $\mathbf{b}_{\text{proj}} \in \mathbb{R}^D$ are learnable projection parameters. The semantic representation $\mathbf{H}^{\text{SEM}} = \{h_1^{\text{SEM}}, \dots, h_T^{\text{SEM}}\} \in \mathbb{R}^{T \times D}$ encodes holistic facial context that enriches local muscle dynamics for robust detection.

2.2.3 Attentive Feature Harmonization Module The physiological and semantic streams provide complementary cues for deepfake detection. The Attentive Feature Harmonization Module integrates these representations through an adaptive attention mechanism that enables physiological features to selectively query semantic context, identifying evidence that supports or contradicts observed muscle dynamics. This attentive interaction reveals inconsistencies between local physiological patterns and global facial context characteristic of synthetic faces.

Adaptive Feature Integration. The attention mechanism first computes query, key, and value projections: $Q = \mathbf{H}^{AU} \mathbf{W}_Q$, $K = \mathbf{H}^{\text{SEM}} \mathbf{W}_K$, $V = \mathbf{H}^{\text{SEM}} \mathbf{W}_V$, where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times d_k}$ are learnable projection matrices and d_k is the key dimension. The scaled attention scores S and normalized weights α are computed as:

$$S = \frac{QK^T}{\sqrt{d_k}} \in \mathbb{R}^{T \times T}, \quad \alpha_{ij} = \frac{\exp(S_{ij})}{\sum_{k=1}^T \exp(S_{ik})} \quad (4)$$

where the scaling factor $\sqrt{d_k}$ stabilizes gradients and α_{ij} represents the attention weight from query position i to key position j . For each of h parallel attention heads with head-specific projections $\mathbf{W}_Q^i, \mathbf{W}_K^i, \mathbf{W}_V^i$, the output is computed as the weighted sum of values, where $\alpha^{(i)} \in \mathbb{R}^{T \times T}$ and $V^{(i)} \in \mathbb{R}^{T \times d_k}$ denote the attention weights and projected values for head i :

$$\text{head}_i = \alpha^{(i)} V^{(i)} \in \mathbb{R}^{T \times d_k} \quad (5)$$

Table 1: Deepfake detection performance on benchmark datasets. Best and second-best results are in bold and underlined, respectively.

Method	FF++		DFDC		DFGC		FakeAVCeleb		HiDF	
	F1↑	AUC↑	F1↑	AUC↑	F1↑	AUC↑	F1↑	AUC↑	F1↑	AUC↑
Effort [15]	0.808	0.887	0.772	0.748	0.922	0.968	0.956	0.895	0.874	0.944
TALL [14]	0.367	0.656	0.581	0.784	0.780	0.647	0.932	0.970	0.889	0.953
SBI [11]	0.813	0.939	0.237	0.661	0.255	0.785	0.814	0.942	0.221	0.662
SLADD [2]	0.668	0.782	0.829	0.916	0.683	0.958	0.818	0.940	0.677	0.942
FTCN [17]	0.828	0.956	0.437	0.809	0.915	0.978	0.877	0.889	0.851	0.941
UIA_ViT [19]	0.726	0.805	0.779	0.910	0.965	0.994	0.942	0.977	0.859	0.924
SIA [12]	0.795	0.872	0.800	0.891	0.739	0.840	0.871	0.908	0.803	0.873
HAUTE(w/o Phys.)	0.875	0.898	0.847	0.887	0.778	0.679	0.927	0.832	0.720	0.843
HAUTE(w/o Sem.)	0.723	0.785	0.664	0.657	0.791	0.709	0.927	0.604	0.652	0.675
HAUTE(all)	0.896	0.963	0.923	0.953	0.973	0.995	0.991	0.992	0.897	0.960

$$\mathbf{H}^{\text{fused}} = \text{LN} \left(\mathbf{H}^{\text{AU}} + \text{Drop} \left(\text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}_O \right) \right) \quad (6)$$

where LN and Drop denote layer normalization and dropout, respectively, $\mathbf{W}_O \in \mathbb{R}^{hd_k \times D}$ projects the concatenated outputs from h heads (each of dimension d_k) to dimension D , with residual connections and layer normalization for training stability.

Video-Level Prediction. The fused sequence $\mathbf{H}^{\text{fused}} \in \mathbb{R}^{T \times D}$ is aggregated via temporal mean pooling to obtain a video-level representation:

$$\mathbf{h} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_t^{\text{fused}} \in \mathbb{R}^D \quad (7)$$

h is passed through a non-linear classification module with ReLU activations and dropout regularization to produce the final detection logit. HAUTE is optimized end-to-end using binary cross-entropy loss with label smoothing:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [\tilde{y}_i \log p_i + (1 - \tilde{y}_i) \log(1 - p_i)] \quad (8)$$

where N , p_i , and $\tilde{y}_i$ denote the batch size, predicted probability, and smoothed label, respectively.

3 Experimental Results

3.1 Experimental Settings

Datasets. We evaluate HAUTE on five widely-used deepfake benchmarks. FaceForensics++ (FF++) [10] contains 4,000 fake videos generated by applying four manipulation methods to YouTube videos. Deepfake Detection Challenge (DFDC)[3] provides 104,500 fake videos as a large-scale challenge dataset created using eight face swap techniques. FakeAVCeleb [6] is an audio-visual deepfake dataset comprising 500 celebrities selected with consideration for racial and gender balance. To assess detection performance against deepfakes generated in real-world scenarios, we additionally employ DeepFake Game Competition (DFGC) [9] and Human-Indistinguishable DeepFake (HiDF) [5], which include data produced using commercial tools.

Implementation Details. The sequence length is set to $T_{\max} = 32$ frames across all experiments. The Temporal Physiological Encoding Stream is configured with hidden size 64 to generate representations of shared embedding dimension $D = 128$. The Holistic Semantic Encoding Stream utilizes CLIP ViT-Large/14 with frozen weights as a pretrained vision encoder. The Attentive Feature Harmonization Module employs $h = 4$ attention heads with dropout rate 0.1, while the classification head uses dropout rate 0.3. The model is trained for 50 epochs using the AdamW optimizer with

Table 2: Cross-dataset evaluation results. FF++/FF++ denotes training, validation, and testing on FF++, while FF++/HiDF denotes training and validation on FF++ and HiDF in a 9:1 ratio and testing on HiDF. Best and second-best results are in bold and underlined, respectively.

Metrics		Effort	TALL	SBI	SLADD	FTCN	UIA_ViT	SIA	HAUTE
FF++ / FF++	F1↑	0.808	0.367	0.813	0.668	0.828	0.726	0.795	0.896
	AUC↑	0.887	0.656	0.939	0.782	<u>0.956</u>	0.805	0.872	0.963
FF++ / HiDF	F1↑	0.798	0.608	0.124	0.701	0.653	0.388	0.604	0.782
	AUC↑	<u>0.878</u>	0.517	0.642	0.827	0.748	0.532	0.750	0.881

learning rate $\eta = 1 \times 10^{-4}$, batch size 32, and label smoothing $\epsilon = 0.1$. All frames are resized to 256×256 resolution. The classification threshold is set to $\tau = 0.5$.

Baselines and Evaluation Protocol. To validate HAUTE’s performance, we compare against seven state-of-the-art deepfake detection methods: Effort [15], TALL [14], SBI [11], SLADD [2], FTCN [17], UIA-ViT [19], SIA [12]. We further conduct ablation studies to analyze the contribution of each component: (1) HAUTE (w/o AU) removes the Temporal Physiological Encoding Stream, (2) HAUTE (w/o Cont.) removes the Holistic Semantic Encoding Stream, (3) HAUTE (all) represents the complete proposed framework. We report F1-score (F1) and Area Under the ROC Curve (AUC) across all experiments.

3.2 Deepfake Detection Performance

Overall Performance. Table 1 compares the proposed method with baseline deepfake detection methods. HAUTE outperforms baseline methods across five benchmarks, achieving high F1-scores and AUC. Notably, strong performance on FakeAVCeleb, which comprises diverse racial and gender groups, demonstrates robustness to demographic variation. The proposed method also excels on commercial tool-based datasets, DFGC and HiDF, highlighting its ability to address deepfakes generated and disseminated in real-world web ecosystems. Furthermore, HAUTE achieves an F1-score of 0.897 and an AUC of 0.960 on HiDF, a high-quality dataset of deepfakes, demonstrating robust detection performance against sophisticated synthetic video.

Ablation Study. Performance changes upon removing each stream clearly demonstrate that the components play distinct yet complementary roles. Removing the Temporal Physiological Encoding Stream degrades performance across all datasets, with DFGC showing a particularly significant drop to 0.679 AUC, a decrease of 0.316 compared to the HAUTE (all). This result indicates that AU dynamics are essential for detecting physiological coordination anomalies in deepfakes, while emphasizing the importance of considering such patterns for detecting deepfakes in real-world web environments. Removing the Holistic Semantic Encoding Stream also results in substantial performance degradation, demonstrating that global contextual information is vital for detecting inconsistencies that cannot be captured solely by local physiological analysis. The consistently superior performance of HAUTE (all) validates the importance of integrating fine-grained physiological patterns with holistic semantic context for robust deepfake detection.

Cross-dataset Evaluation. Table 2 evaluates adaptability to limited target domain data by training and validating on FF++ and HiDF in a 9:1 ratio and testing solely on HiDF. This configuration assesses whether models trained on traditional synthesis methods

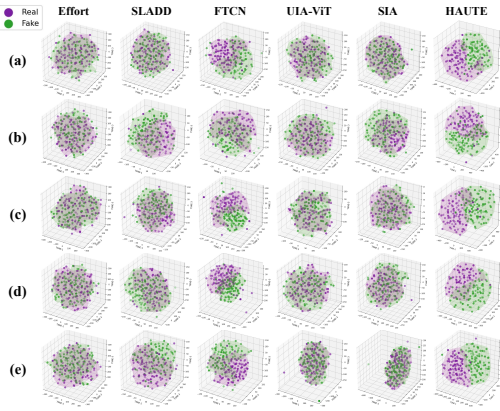


Figure 2: t-SNE visualization of learned representations from HAUTE and top 5-performing baselines across five datasets. (a) FF++, (b) DFDC, (c) DFGC, (d) FakeAVCeleb, (e) HiDF. Purple: real samples, green: fake samples.

can effectively detect sophisticated deepfakes in real-world web environments with minimal high-quality samples. HAUTE achieves 0.881 AUC and 0.782 F1-score, demonstrating competitive performance with Effort at 0.878 AUC while significantly outperforming remaining baselines, validating superior adaptability with limited target domain data. Such performance indicates that HAUTE can effectively adapt to emerging deepfake techniques in web environments where acquiring large-scale labeled data remains a practical challenge.

3.3 Qualitative Analysis

Figure 2 shows t-SNE visualizations of learned representations from HAUTE and the top 5-performing baseline methods projected into three-dimensional space. HAUTE exhibits distinct clusters between real and fake samples across all five datasets. Notably, HAUTE maintains sharp boundaries between the two classes even on commercial tool-based high-quality deepfakes such as HiDF, whereas most baseline methods exhibit substantial overlap and mixed distributions. These results demonstrate that HAUTE learns highly discriminative representations by integrating physiological dynamics and semantic context, possessing robust properties that effectively distinguish real from fake across diverse synthesis methods and quality levels.

4 Concluding Discussion

In this paper, we propose Harmonizing Action Units with Temporal-contextual Embeddings (HAUTE), which integrates physiological dynamics with global semantic context through adaptive attention mechanisms to achieve robust deepfake detection across diverse synthesis methods and quality levels. Extensive experiments demonstrate that HAUTE achieves superior performance and strong adaptability compared to existing methods. Specifically, robust detection performance on commercial tool-based high-quality deepfakes underscores HAUTE’s ability to address sophisticated synthetic content disseminated in real-world web environments. Moreover, the encoder-agnostic design of the Holistic Semantic Encoding Stream enables seamless integration with emerging vision models, ensuring scalability and adaptability to the proliferation of new manipulation methods across web environments.

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